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A

PROJECT REPORT ON

Automated Side Mirror Car Wiper System

(SDG 9: Industry, Innovation and Infrastructure)

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE IN THE PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE OF

BACHELOR OF ENGINEERING IN ELECTRONICS AND TELECOMMUNICATION

BY

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UNDER THE GUIDANCE OF

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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

BY ALL INDIA SHRI SHIVAJI MEMORIAL SOCIETY'S INSTITUTE OF INFORMATION TECHNOLOGY, PUNE - 411001

ACADEMIC YEAR: 2024-25

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Department of Electronics and Telecommunication Engineering

CERTIFICATE

This is to certify that Project Report entitled

Automated Side Mirror Car Wiper System

(SDG 9: Industry, Innovation and Infrastructure)

Submitted By

Shraddha Kapse (B400250180)
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Prasad Vhanmane (B400250193)

is the record of bonafide work carried out by them in partial fulfillment of the requirement for the Award of the Degree of **Bachelor of Engineering (Electronics and Telecommunication)**, as prescribed by the Savitribai Phule Pune University in the Academic Year 2024-2025.

This project report has not been earlier submitted to any other Institute or University for the award of any degree.

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The motivation factor for this work was the inspiration given by our honorable principal **Dr. P. B. Mane.**

Lastly, I am thankful to those who have directly or indirectly supported for our work.

Sign	Sign	Sign
Shraddha Kapse	Pranav Madane	Prasad Vhanmane

Academic Year: 2024-25

Vision

To provide quality education in Electronics & Telecommunication engineering with professional ethics.

Mission

To develop technical competency, ethics for professional growth and a sense of social responsibility among students.

Program Educational Objectives

1. To provide students with solid foundation in mathematics and fundamentals of Electronics & Telecommunication engineering.

2. To prepare students to analyze and solve real life problems.

3. To inculcate professional and ethical attitude and good communication skills among the students.

4. To develop social awareness among the students.

Program Specific Outcomes

Graduates will be able to

1. Apply fundamentals and knowledge of electronics, telecommunication and mathematics to analyze, design and develop electronics and telecommunication systems.

2. Select different software tools, test and measurement equipment and use them efficiently for system solutions.

Program Outcomes

1. Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems. .**[Engineering Knowledge]**

2. Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the firstprinciples of mathematics, natural sciences, and engineering sciences. **[Problem Analysis]**

3. Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations. [**Design/development of solutions]**

4. Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. [**Conduct investigations of complex problems]**

5. Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations. **[Modern tool usage]**

6. Apply reason informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice. [**The engineer and society]**

7. Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. [**Environment and sustainability]**

8. Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice. [**Ethics]**

9. Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. [**Individual and team work]**

10. Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[Communication]**

11. Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments. [**Project management and finance]**

12. Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. [**Life-long learning]**

ABSTRACT

In rainy conditions, maintaining clear visibility through side windows is essential for safe driving, especially when checking blind spots or turning. However, raindrops and water streaks can obscure a driver's view. To address this, the "Automated side mirror car wiper system" provides an innovative solution by automatically cleaning the side windows during rain.

This project proposes an Automated Side Mirror Car Wiper System to address this limitation. The system integrates sensors, a microcontroller, and a compact wiper mechanism to detect moisture and automatically activate a wiper for the side mirrors. A rain sensor or similar moisture-detection technology triggers the system to ensure clear visibility in real time. The system can operate independently or synchronize with the car's existing Side Mirror wipers.

The "Automated side mirror car wiper system" is particularly useful in heavy rain or when vehicles are moving at slower speeds, where natural wind flow may not be sufficient to clear the windows. Designed to work not only during rainy conditions, this system enhances safety by allowing drivers to maintain clear visibility, checking side mirrors, contributing to overall driving comfort and reducing accident risks.

The system consists of key components, including a rain sensor to detect moisture or precipitation, an Rain sensor for detection, and a microcontroller to process the sensor inputs and control the system. A miniaturized motor powers the wiper mechanism, designed specifically for compact and efficient operation on side mirrors. Additional components, such as power supply unit, and mounting hardware, ensure seamless functionality and integration with the vehicle. We can’t say that the system is synchronized with the car’s existing wipers or operates autonomously based on the detected conditions.

By automating the cleaning process, this system enhances safety, particularly during inclement weather or in situations requiring frequent mirror checks, such as urban driving or parking. It also offers a cost-effective and energy-efficient solution that can be integrated into both new and existing vehicles.

During experimental testing, the system's sweeping mechanism was evaluated under varying rain intensities. The arc length of the wiper motion remained approximately constant (0.314m) throughout the tests. As rain intensity increased, the accumulation time on the rain sensor decreased, leading to a corresponding drop in the sensor's output voltage. At a low rain intensity (3.2 V output), the sweep time was recorded as 10.0 seconds, resulting in a speed of 0.0314 m/s. With increasing rain intensity and a sensor output of 2.6 V, the sweep time reduced to 9.0 seconds, and the speed increased to 0.0349 m/s. At 2.0 V, the sweep time was 8.0 seconds with a speed of 0.0393 m/s. Further increase in rain intensity resulted in a 1.4 V output, reducing sweep time to 7.0 seconds and increasing speed to 0.0419 m/s. At the highest intensity, the sensor output dropped to 0.8 V, sweep time decreased to 6.0 seconds, and the speed reached 0.0523 m/s. These results show that the system effectively adjusts the wiper speed in real time based on rain intensity, using the sensor's voltage output as a reliable input signal.

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CHAPTER 1:

INTRODUCTION

INTRODUCTION

The "Automated Side Mirror Car Wiper System" concept intends to improve driving efficiency and vehicle safety by resolving a frequent problem that arises during periods of intense precipitation: poor sight through the side windows. Conventional wiper systems are usually made for the front and rear Side Mirrors, leaving the side windows vulnerable to water buildup. This can make it difficult for drivers to see clearly and raise the chance of accidents, particularly in inclement weather.

In order to maintain clear vision, this proposal presents an intelligent, automatic wiper system for automobile side windows that will activate in reaction to rain. A servo motor, a wiper mechanism, and a microcontroller board make up the system. As the system's brain, the microcontroller processes data from rain sensors and manages the servo motor to run the wipers in real time. Water is effectively removed from the side pane by the wiper function, guaranteeing the driver always has an unhindered vision.

The concept greatly lessens the need for manual intervention by automating the side mirror wiper system, improving driver convenience and safety. Distractions are reduced and drivers are assisted in keeping their attention on the road by the real-time reaction to changing weather conditions. Additionally, by increasing visibility in severe rain, this technology lowers the chance of accidents and promotes efficient driving. An advancement in automotive technology, this automated solution guarantees safer driving and increased dependability in inclement weather.

A car wiper is a device which is used to remove droplets of rainwater from a Side Mirror. Nowadays, each and every vehicle is provisioned with the wiper to avoid the accidents and to decrease the human intervention in controlling the wiper to ensure luxury. A wiper generally consists of a metal arm and a long rubber blade.

CHAPTER 2:

LITERATURE REVIEW

LITERATURE REVIEW

2.1 LITERATURE REVIEW

Moni Sree, Dharshana N, Dheekshitha S, Archana Presented A paper titled “Automatic rain sensing wiper system using a 555 timer” in article [1] . System activates a wiper through a sensor-based detection mechanism without manual switching, enhancing driver safety by reducing distraction. The circuit is divided into four main components: a 555 IC in astable mode, a comparator, motor driver circuitry, and a rain detector. Their design demonstrates the conversion of manual to automatic operation, targeting cost-efficiency and reliability in rain detection.

In [2] Vinayak Sagare, D. N. Sonawane, Mukul Joshi, M. A Joshi developed a resistive rain sensor with a linearized output using a customized PIC microcontroller. Their work, titled “A novel and cost-effective resistive rain sensor for automatic wiper manage: circuit modeling and implementation”, emphasizes efficiency by minimizing the non-linearity in sensor output through circuit-based linearization.

In [3] Lubna Alazzawi, Avik Chakravarty introduced a reconfigurable computerized rain-sensitive Side Mirror wiper system. Their paper “Design and implementation of a reconfigurable computerized rain sensitive Side Mirror wiper” highlights fuzzy logic control and PWM for motor actuation, ensuring adaptability and precision.

In [4] P. Abhilash Reddy, G. Sai Prudhvi, P. J. Surya Sankar Reddy, Dr. S. S. Subashka Ramesh proposed an Arduino-based automatic rain sensing car wiper system to enhance driver safety by eliminating manual intervention during rainfall. The system uses a rain sensor, LCD display, servo motor, and LM393 comparator, interfaced with an ATmega8 microcontroller. When moisture exceeds a threshold, the sensor triggers the Arduino, which processes the data and activates the wiper motor. The LCD module displays rainfall intensity and wiper speed, allowing real-time feedback to the driver. This system is intelligent, starts automatically during rain, and stops once the rain ends, thereby reducing driver distraction and enhancing road safety.

In [5] Akila Wijesinghe proposes an automated rain wiper system aimed at enhancing driver visibility during rainy conditions without manual intervention. The system automatically adjusts the speed of the wipers based on rainfall intensity using a rain sensor and microcontroller (PIC16F877A). It comprises components such as the L298D motor driver, L7805 voltage regulators, and a rain drop sensor to detect precipitation levels. The microcontroller processes sensor data to control wiper speed via PWM signals. This design is low-cost, energy-efficient, and suitable for standard vehicles, promoting road safety by reducing driver distractions. The research demonstrates a practical, cost-effective solution for rain-sensing automation in automobiles.

In [6] Prajapati and Khatri present a system titled “Rain Operated Automatic Wiper in Automobile Vehicle”, focused on enhancing driver safety and convenience by automating Side Mirror wiper operation without manual input. The system detects rainfall using sensors and activates the wipers accordingly, allowing drivers to concentrate on the road. It aims to eliminate the need for drivers to manually adjust wiper speed, especially during varying rain intensities. The proposed design employs an impedance and IR sensor-based rain detection mechanism connected to a microcontroller, which processes the input and activates a motor control logic for appropriate wiper speed. The data processing unit is built around an AVR Atmega8 microcontroller, selected for its high analog input capability. The system logic includes initialization, rain sensing, and speed regulation via relays, based on rainfall intensity. This project targets both retrofit and new vehicle applications to reduce accidents and improve vehicle automation using a cost-effective and modular design approach.

In [7] Devi et al. present a project titled "Automatic Rain Sensing Wipers Using Arduino," aimed at enhancing driver safety by automating Side Mirror wiper operation based on rainfall intensity. The system utilizes an Arduino Uno, a rain sensor, a servo motor, and an LCD module to detect precipitation and adjust the wiper speed accordingly. The proposed design eliminates the need for manual intervention, thus minimizing driver distraction and potential accidents during rainfall. The block diagram shows a four-stage system where rain is detected, processed, and used to control a servo motor that operates the wiper. The rain sensor functions based on the resistance change caused by water droplets, and the Arduino processes this signal to determine motor speed. This low-cost, efficient automation model is particularly suited for use in modern vehicles to improve road safety and driving comfort during adverse weather conditions.

In [8] the article Tim Brophy, Darragh Mullins , Ashkan Parsi, Jonathan Horgan, Enda Ward, Patrick Denny, Ciarán Eising, Brian Deegan, Martin Glavin, and Edward Jones, presents a comprehensive review of how adverse weather conditions, particularly rain, affect the performance of visible spectrum cameras used in automated vehicles for tasks like object detection, classification, avoidance, and path planning. The authors introduce a data communication-inspired "Image Formation Framework", which maps the flow of visual information from real-world objects through environmental channels (like rain) to the sensor (camera), and finally through the processing pipeline. This model helps to systematically analyze where and how rain impacts image data quality and perception reliability.

In [9] paper by Saqlain Ahmed discuss the design and development of an Automatic Rain Operated Wiper system, which uses a conduction (touch) sensor, a control unit, and a wiper motor to automatically activate windshield wipers during rainfall. The system is powered by a battery and utilizes a PIC16F73 microcontroller to process signals from the sensor and control the wiper motor accordingly. When rain is detected on the vehicle glass, the sensor sends a signal to the control unit, which then turns on the motor without requiring driver intervention. The project highlights its applicability in four-wheelers and its reliability for industrial and domestic use, demonstrating how embedded systems can automate essential vehicle functions for enhanced safety and convenience.

2.2 SUMMARY OF LITERATURE REVIEW:

The reviewed literature presents a range of approaches for designing and implementing automatic Side Mirror wiper systems aimed at enhancing driver safety, comfort, and visibility during rainfall. Traditional manually operated systems are considered inconvenient and distracting, prompting the development of automated alternatives that activate wipers based on real-time environmental sensing. Most studies utilize moisture or rain sensors—such as resistive, capacitive, optical, or IR-based types—to detect precipitation on the mirror surface. These sensors are interfaced with microcontrollers like PIC, Arduino, or ATmega, which process the sensor signals and control wiper actuation through motors, actuators, or servo mechanisms. Some systems integrate advanced features such as fuzzy logic, PWM motor control, and real-time speed adjustments based on rainfall intensity.

Cost-effectiveness and circuit simplicity are common design goals, with components like 555 timers, LM393 comparators, and L298D motor drivers frequently used to minimize complexity while maintaining reliability. Certain designs also include features such as lifting the wiper blade when the engine is turned off to prevent rubber damage from heat, particularly in hot climates. Other innovations include wireless activation via Bluetooth and the use of LCD displays for real-time feedback. Research also explores sensor output linearization to ensure accurate and consistent performance in varying weather conditions. Overall, these systems aim to eliminate manual intervention, reduce driver distraction, and offer a practical, low-cost solution for integrating rain-sensitive automation into both new and existing vehicles

2.3 DRAWBACKS:

From the literature review following limitations were observed:

- The system may struggle with sensor accuracy in extreme weather conditions, such as heavy snow or fog, which can lead to delayed or incorrect wiper activation.
- Continuous operation of sensors, wipers, and heaters may lead to increased power consumption, potentially impacting the vehicle's battery life, especially in electric vehicles.
- The usage of sensor (rain) may increase the overall cost and complexity of the system, making it less accessible for budget vehicles.
- The system might not be able to adapt to all types of mirrors or vehicle models, particularly with non-standard mirror shapes or unconventional designs.

CHAPTER 3:

PROBLEM STATEMENT

PROBLEM STATEMENT

Maintaining clear visibility through side mirrors during rainy conditions is critical for safe driving, particularly for lane changes, parking, and reversing. However, existing vehicle systems lack dedicated mechanisms to automatically detect and remove rainwater, fog, or debris from side mirrors. Drivers often depend on natural airflow or manual cleaning, which are ineffective during heavy rain or at low vehicle speeds, compromising safety and convenience.

The project aims to design and develop a reliable, cost-effective, and automated side mirror car wiper system that addresses these limitations. The system should ensure real-time cleaning of side mirrors during rainy conditions, integrate seamlessly with vehicle operations, and include essential safety and control features. This solution aims to enhance visibility, driving safety, and convenience in adverse weather conditions.

CHAPTER 4:

AIM & OBJECTIVE

AIM & OBJECTIVE

4.1 AIM

The aim of the project is to design and develop an automated side mirror car wiper system that ensures clear visibility by detecting and removing water, dust, or debris from vehicle side mirrors in real time. This system seeks to enhance driving safety, especially during adverse weather conditions, by integrating compact wiper mechanisms, sensors, and a microcontroller for precise and automated operation. The project aims to provide a cost-effective, reliable, and energy-efficient solution suitable for both modern and existing vehicles, minimizing driver intervention and contributing to safer and more comfortable driving experiences.

4.2 OBJECTIVES

- To ensure clear visibility for drivers during heavy rain.
- To provide an affordable solution for side mirror wiper systems.
- To reduce the risk of accidents by maintaining driver visibility in adverse weather conditions.

4.3 SPECIFICATIONS

- **Wiper Motor Speed:** 50–100 RPM
- **Detection Range:Rain/Moisture Sensor:** Sensitivity to moisture levels as low as 0.1 mm of precipitation**.**
- **Accuracy:** 90%
- **Reaction Time:** Less than 1 second from detection to activation
- **Power Supply:**12V DC
- **Wiper Sweep Area:** Covers 90% of the side mirror surface
- **Durability:** Weather-resistant components to withstand rain, snow, and dust.

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CHAPTER 5:

SOCME

SYSTEMS THINKING-SOCME for Multidisciplinary Approach of Problem Solving

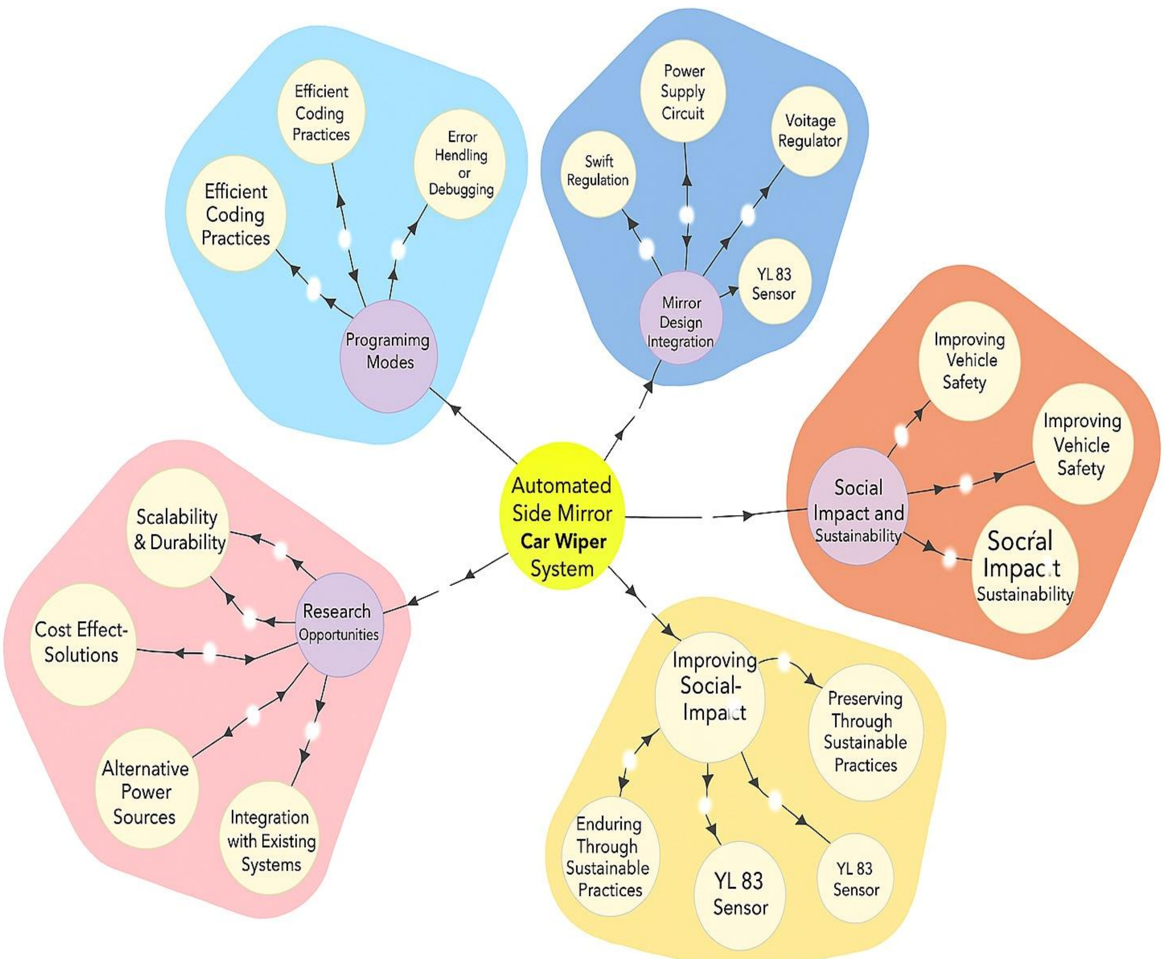

Figure 1 : SOCME Diagram

CHAPTER 6:

METHODOLOGY

METHODOLOGY

1.Requirement Analysis

- Identify the need for real-time cleaning of side mirrors during rainy or dusty conditions.
- Determine the components needed: rain sensor, motor, microcontroller (STM32), and user interface.

2.System Design

- Develop a block diagram to represent the system components and their interconnections.
- Create a circuit diagram for hardware components, including sensor connections, motor driver circuits, and the microcontroller interface .

3.Component Selection

- Select a rain sensor capable of detecting moisture or precipitation with high sensitivity.
- Choose a motor with sufficient torque to operate the compact wiper mechanism.
- Use STM32 as the microcontroller for processing sensor data and controlling the wiper motor.

4.Hardware Integration

- Connect the rain sensor to the microcontroller for real-time moisture detection.
- Interface the motor with the wiper mechanism to clear the side mirror surface.
- Add a manual override switch for user control and an LED indicator for system status

5.Firmware Development

- Write code for the microcontroller (STM32) microcontroller to process sensor inputs, execute the control algorithm, and drive the motor.
- Integrate features for manual control and synchronization with the vehicle's wiper system.

6.Testing

- Test individual components (rain sensor, motor, wiper mechanism, etc. for proper functionality.
- Adjust motor speed and wiper sweep area to ensure complete cleaning of the side mirror performance.

7. Validation

- Test & Validate the system under environmental condition (Rain) to ensure reliability.

CHAPTER 7:

HARDWARE DESIGN

HARDWARE DESIGN

7.1 BLOCK DIAGRAM & EXPLANATION

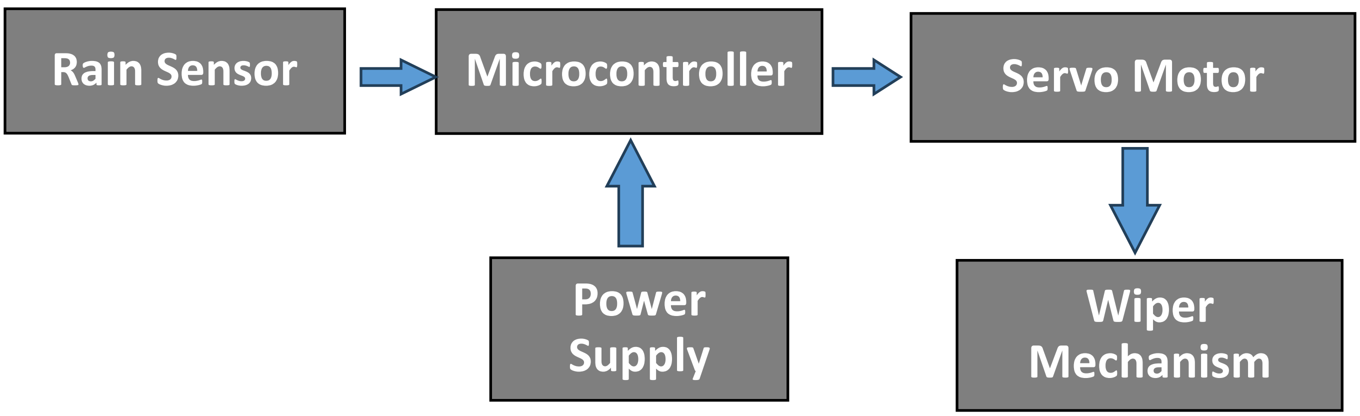

Figure 2: Block Diagram

Description of Block Diagram

Figure 2 shows the block diagram of "Automated car wiper system" . The system is designed to automate the wiper mechanism based on rainfall intensity using an STM32 microcontroller. A rain sensor continuously monitors precipitation levels and generates an analog voltage signal proportional to the rain intensity. This signal is fed into the STM32 microcontroller, which processes the input and determines the appropriate speed for the wiper operation. The microcontroller then sends control signals to a servo motor, which adjusts the movement of the wiper mechanism accordingly. The entire system is powered by a regulated power supply that ensures consistent operation of both the microcontroller and motor components

1.Rain Sensor:

- The YL-83 rain sensor module consists of a rain detection board and a comparator circuit.
- It detects the presence of raindrops through conductivity changes when water touches the sensor surface.

- The analog or digital output from the sensor indicates the intensity of rainfall and is sent to the microcontroller for processing.

2.Microcontroller (STM32):

- The STM32 microcontroller serves as the core processing unit of the system.
- It receives input from the rain sensor and interprets the signal to determine whether rain is present and its intensity.

3.Servo Motor (SG90):

- The SG90 servo motor acts as the actuator for the wiper mechanism.
- Controlled by PWM signals from the STM32 microcontroller, the servo adjusts the angle and motion of the wiper.

4.Power Supply:

- The power supply unit provides the necessary operating voltages (typically 5V and 3.3V) for the STM32 microcontroller, rain sensor, and servo motor.
- It ensures stable power delivery to maintain consistent operation of all system components.

5.Signal Measurement and Processing:

- The wiper mechanism physically clears water from the Side Mirror using mechanical motion driven by the servo motor.
- It is linked directly to the servo, translating motor movements into linear or rotary wiping motion.
- The system operates automatically, turning on and adjusting speed or motion based on real-time rainfall detection

Figure 3 shows the connection diagram of hardware components

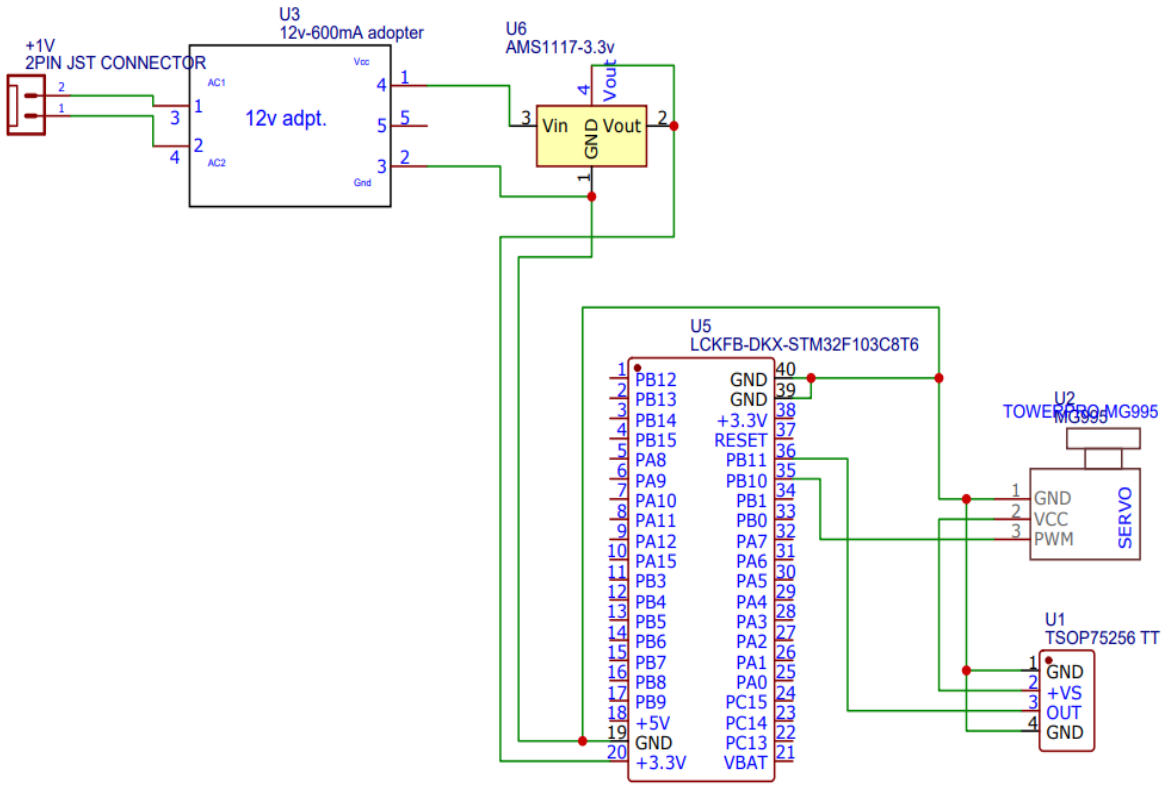

Figure 3: Connection Diagram

7.2 HARDWARE COMPONENTS

- **Microcontroller (STM32)**
- **Servo Motor (SG90)**
- **Rain Sensor (YL83)**
- **Rain Sensor Comparator**
- **Power Supply**
- **Voltage Regulator (7812)**
- **ST Link V2 Programmer**
- **Zero PCB**

1. Microcontroller (STM 32)

The **STM32** is a family of 32-bit microcontrollers developed by STMicroelectronics, based on the ARM Cortex-M core. Known for its high performance, low power consumption, and rich peripheral set, STM32 is widely used in embedded systems and real-time applications, including automation and sensor-based control.

Features:

- **Core**: ARM Cortex-M (varies by model; e.g., Cortex-M0, M3, M4, M7).
- **Clock Speed:** Up to 72 MHz (STM32F103) or higher depending on the variant.
- **Flash Memory**: Ranges from 16 KB to 2 MB.
- **RAM**: Up to 512 KB (typically 20–96 KB for STM32F1/F3 series).
- **GPIO**: Multiple General Purpose I/O pins with alternate functions.
- **Communication Interfaces**: I2C, SPI, UART, CAN, USB, and more.
- **Timers**: Multiple advanced timers, PWM generation support.
- **ADC/DAC**: Built-in analog-to-digital and digital-to-analog converters.
- **Power Efficiency**: Low-power modes for battery-operated systems.
- **Debugging**: Supports SWD and JTAG for debugging and flashing.

Applications:

- Sensor and actuator control systems.
- Medical and consumer electronics.
- Automotive applications.
- Real-time data processing and embedded control.

Technical Specifications:

Table 1: STM 32 Technical Specification

Specification	Details
Core	ARM Cortex-M3
Operating Voltage	2.0V to 3.6V
Clock Speed	Up to 72 MHz
Flash Memory	64 KB
SRAM	20 KB
GPIO Pins	37 (up to 37 I/Os depending on configuration)
Communication Interfaces	2x I2C, 3x USART, 2x SPI, CAN, USB
Power Consumption	Low-power modes supported

2. Servo Motor (SG-90)

The SG90 is a lightweight, low-cost servo motor commonly used in hobby electronics, robotics, and DIY automation projects. It offers precise position control within a limited rotation range and is ideal for small-scale mechanisms.

Features:

- **Operating Voltage:** 4.8V to 6V
- **Stall Torque:** 1.8 kg·cm @ 4.8V
- **Speed:** ~0.1 sec/60° @ 4.8V
- **Control Signal:** PWM (Pulse Width Modulation)
- **Rotation Range:** 0° to 180°
- **Weight:** Approx. 9 grams
- **Connector:** 3-wire (VCC, GND, Signal)
- **Material:** Plastic gears (in standard SG90)

Applications:

- Mini robotic arms
- RC planes and cars
- Automatic wipers
- Positioning systems in embedded projects
- Pan-tilt camera systems

- **Pin Description:**

Table 2: Servo Motor (SG-90)

Pin	Name	Function	Signal/Voltage	Description
1	**VCC**	Power Supply	5V DC	Supplies power to the Rain sensor.
2	**GND**	Ground	0V	Connects to the ground of the power supply to complete the circuit.
3	**PWM**	Control signal input	TTL Level	Receives the PWM signal from the microcontroller to control the shaft position.

3. Rain Sensor

The YL-83 is a rain detection sensor module designed to detect the presence and intensity of rainfall. It is commonly used in weather monitoring systems and automated systems for controlling actuators based on rain detection.

Features:

- Detects water droplets and moisture on the sensor surface.
- Adjustable sensitivity using onboard potentiometer.
- Provides both digital and analog outputs.
- Compact and easy to interface with microcontrollers.
- Operates at low power consumption.

Applications:

• Automatic Side Mirror wiper control in vehicles.

• Smart irrigation and weather stations.

• Rainwater collection and management systems.

• Outdoor equipment protection and automation.

Technical Specifications:

Table 3: Rain sensor Technical Specification

Specifications	Details
Operating Voltage	3.3V – 5V DC
Detection Range	Detects Presence of Droplets
Output signal	Digital (0 or 1) and Analog (variable voltage)
Sensor Type	Resistive rain detection
Interface	3-pin: AO (Analog), DO (Digital), GND
Sensor Plate Size	~5.4 cm x 4.0 cm
Operating Temperature	-40°C to 80°C
Applications	Rain detection, smart wipers, weather stations

4. Rain Sensor Comparator Module

The rain sensor comparator module is used to detect water droplets on the rain sensor board and generate corresponding analog and digital signals for microcontroller interfacing. It plays a key role in automating systems like wipers, window controls, and weather-based monitoring.

Features:

• Operates on 3.3V to 5V DC.

• Adjustable sensitivity using onboard potentiometer.

• Dual output: Analog (moisture level) and Digital (threshold-based).

• Built-in LM393 comparator for digital signal generation.

Applications:

• Rain-activated smart systems.

• Automated Side Mirror wipers.

• Weather station modules.

Technical Specifications:

Table 4: Rain Sensor Comparator Specifications

Specifications	Details
Comparator Chip	LM393 Voltage Comparator
Operating Voltage	3.3-5v Dc
Digital Output	TTL Logic (0V or 5V)
Analog Output	Varies with moisture level on sensor
Adjustable Sensitivity	Yes, via onboard potentiometer
Indicator LED	Onboard LED Turns On
Interface Pins	AO (Analog), DO (Digital), VCC, GND
Application	Rain detection, smart wipers, weather stations

5. Power Supply

A 12V AC to DC power supply unit is used to convert mains AC voltage (110/220V) into a stable 12V DC output. It is essential in electronic circuits that require regulated low-voltage DC power for sensors, actuators, and microcontrollers.

Features:

• Converts 110V/220V AC to 12V DC.

• Provides regulated and stable output.

• Short-circuit and overload protection.

• Compact and durable design.

Applications:
- Powering microcontroller-based systems.
- Home automation devices.
- Sensor modules and relay boards.

Technical Specifications:

Table 5: Power Supply Specifications

Specifications	Details
Input Voltage	110-220v
Output Voltage	12V DC
Output Current	Typically 1A-5A
Output Type	Regulated DC
Protection Features	Short-circuit, overload, over-voltage
Operating Temperature	0°C to 50°C
Efficiency	≥85%
Application Use	General-purpose DC power supply

6. Voltage regulator (7812)

A 7812 voltage regulator is a fixed linear voltage regulator that outputs a constant 12V DC, commonly used in power supply circuits to ensure steady voltage for components and systems.

Features:
- Fixed 12V DC regulated output.
- Built-in thermal shutdown and short-circuit protection.
- Reliable and easy to use in linear regulator circuits.

Applications:

• Microcontroller power supplies.

• Analog circuit regulation.

• Embedded system voltage control.

Technical Specifications:

Table 6: Voltage regulator (7812) Specification

Specifications	Details
Output Voltage	12V
Input Voltage	14-35 V
Output Current	Upto 1.5A
Protection Features	Thermal overload & short-circuit
Package Type	TO-220

7. ST Link V2 programmer

The ST-Link V2 is an in-circuit debugger and programmer for STM8 and STM32 microcontroller families. It provides a reliable connection between a host computer and a target MCU for firmware programming and debugging.

Features:

• Supports STM8 and STM32 MCUs.

• USB interface for power and communication.

• Compact and robust design.

Applications:

• Microcontroller programming.

• Embedded system debugging.

• Firmware development and testing.

Technical Specifications:

Table 7: ST Link V2 programmer Specification

Specifications	Details
Supported MCUs	STM8 and STM32
Interface Type	USB 2.0
Connector Type	4-pin SWIM & 10-pin SWD
Programming Voltage	3.3V and 5V (selectable)
Compatibility	Windows/Linux IDEs

7.3 WORKING

- The Side Mirror Wiper System is designed to automatically activate and control a wiper mounted on a vehicle's side mirror based on real-time weather conditions. The system ensures improved visibility during rain without requiring manual intervention.
- The rain sensor continuously monitors the presence of water droplets on the surface. When rain is detected, the sensor sends a digital signal to the STM32 microcontroller.
- The STM32 processes the input from the sensor using a comparator circuit, which distinguishes between dry and wet conditions by comparing signal levels.
- Upon detecting rain, the STM32 sends a PWM (Pulse Width Modulated) control signal to the servo motor, which drives the wiper mechanism.
- The servo motor is responsible for moving the wiper across the mirror in a programmed sweep pattern. It operates within a set angular range, ensuring complete coverage of the mirror surface.
- Once the rain stops and the sensor detects a dry condition again, the STM32 signals the servo motor to stop operation, conserving energy and preventing unnecessary wear.
- This fully automated system requires no user interaction and ensures that the side mirror remains clear during rainy weather, enhancing safety and driver visibility.

7.4 SYSTEM ARCHITECTURE

- The rain sensor detects rain intensity and sends a corresponding voltage signal to the STM32 microcontroller.
- The STM32 processes this signal and generates a PWM signal to control the SG90 servo motor. The servo motor, powered by a 5V supply, drives the wiper mechanism on the side mirror.
- The STM32 is powered through a 3.3V regulator (AMS1117) from a 12V adapter. Ground is shared among all components for proper reference.

CHAPTER 8:

SOFTWARE DESIGN

SOFTWARE DESIGN

8.1 FLOWCHART

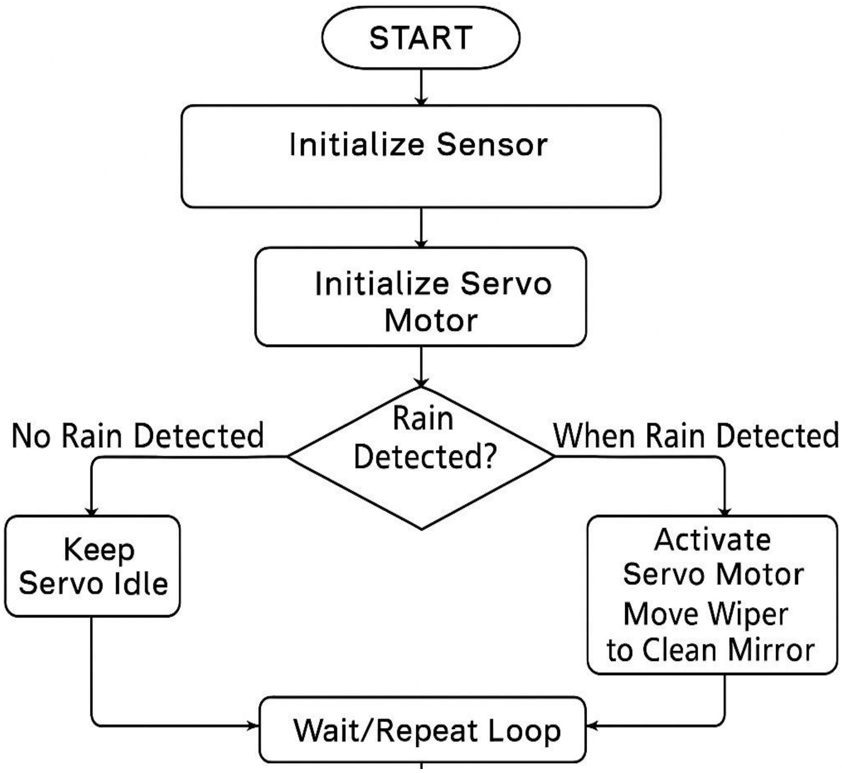

Figure 4: Software Flowchart

Explanation of software flowchart shown in figure 4:

1. START:

- The system is powered ON and begins execution of the control logic for the wiper.

2. Initialize Sensor:

- The rain/moisture sensor (e.g., resistive or capacitive type) is initialized by the microcontroller to start monitoring environmental conditions.

3. Initialize Servo Motor:

- The servo motor, which is mechanically linked to the side mirror wiper, is initialized and made ready to receive control signals.

4. Rain Detected? (Decision Block):

- The sensor checks for the presence of water or rain on the mirror surface.
- The system enters a decision-making process

(a) If rain is NOT detected:

- Proceed to Step 5.

(b) If rain IS detected:

- Proceed to Step 6.

5. No Rain Detected – Keep Servo Idle:

- If the sensor does not detect any rain, the servo motor remains in idle state.
- No motion is triggered, preserving power and mechanical life.

6. When Rain Detected – Activate Servo Motor:

- When rain is detected, the microcontroller sends a signal to the servo motor.
- The motor activates and moves the wiper across the mirror to clean the water droplets.

7. Wait / Repeat Loop:

- After either state (idle or wipe action), the system returns to the monitoring loop.
- The sensor continuously checks for rain, allowing real-time automated wiper control.

8.2 USED SOFTWARE

1. STM32CubeIDE:

STM32CubeIDE is an integrated development environment (IDE) developed by STMicroelectronics for programming STM32 microcontrollers.

Key Features:

- Combines code editing, compilation, and debugging in one platform.
- Built-in support for STM32 microcontrollers.
- Supports C and C++ programming.
- Integrated STM32CubeMX for peripheral configuration and code generation.

Applications:

1. Embedded Systems Development – Used in industrial controllers, smart appliances, and automation systems.
2. IoT Devices – Powering smart sensors, home automation, and connected wearables.
3. Robotics – Controls motors, sensors, and communication in robotic systems.
4. Medical Devices – Used in portable diagnostic tools and monitoring equipment.
5. Consumer Electronics – Found in drones, cameras, gaming accessories, and audio systems.

8.3 SIMULATION

In this project, Proteus Design Suite was used to simulate and validate the automated side mirror wiper system before actual hardware implementation. The simulation included key components such as the STM32 microcontroller, rain sensor, SG90 servo motor, and power supply circuitry. Using Proteus, we:

- Created the schematic to ensure correct wiring and pin configuration between all components.
- Simulated varying rain sensor outputs to observe how the STM32 generated PWM signals accordingly. Tested the logic that triggers servo motor movement based on rain intensity thresholds.

- Visualized the motor response in real time, ensuring the wiper activates accurately with changing sensor input.
- Identified and resolved circuit-level issues during simulation, minimizing the risk of hardware damage.

The figure 5 , below illustrates the Proteus simulation setup, demonstrating the real-time functioning of the side mirror wiper system in response to rain detection.

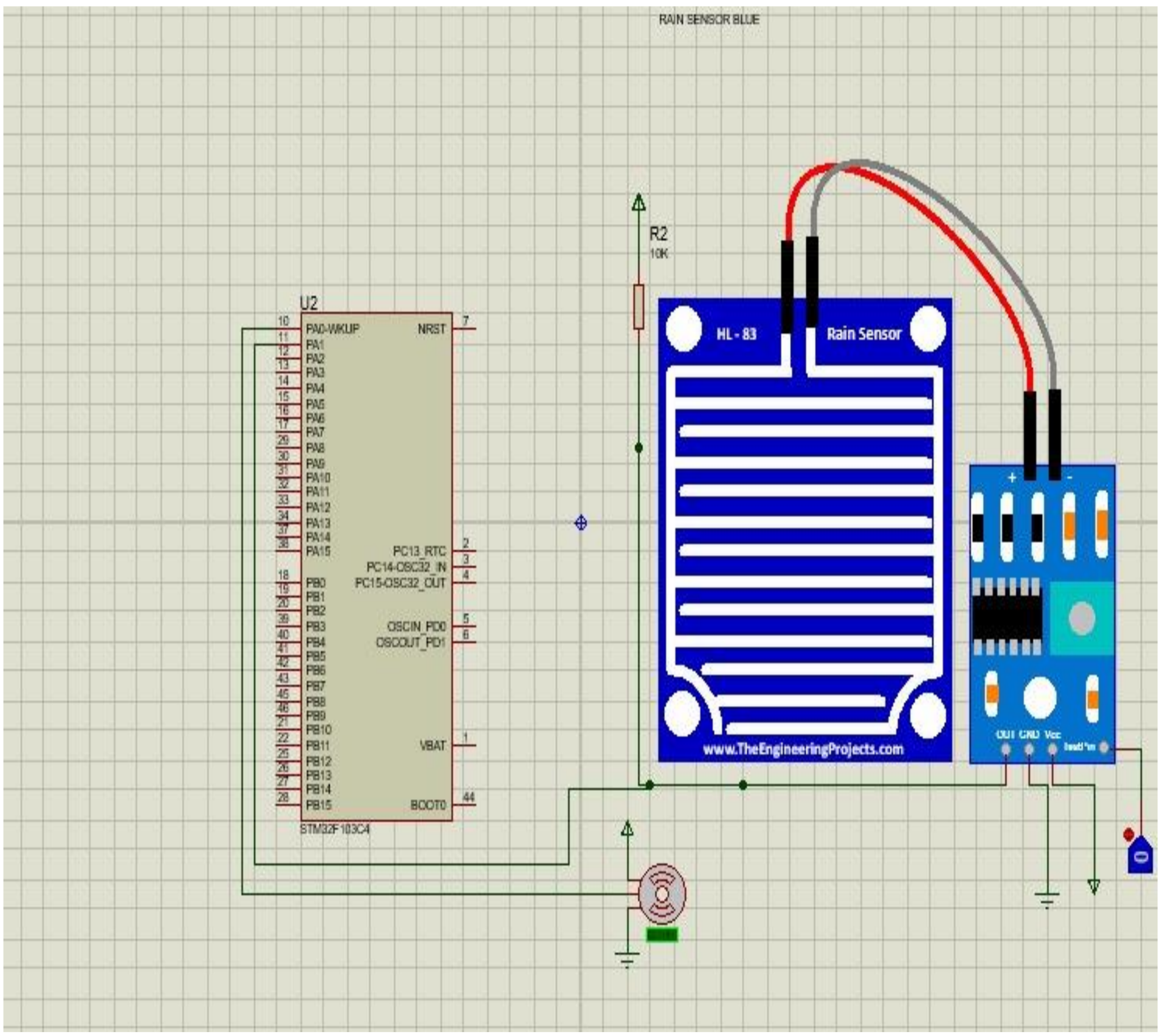

Figure 5: Screenshot of Software Simulation

CHAPTER 4:

RESULTS

RESULTS

The developed system demonstrated accurate speed control, quick responsiveness, robustness under varying conditions, and good overall efficiency. The observed actual results are outlined below:

1. **Reliable Water Detection:** The YL-83 rain sensor is designed to accurately detect the presence of water or raindrops on the side mirror surface. This enables timely activation of the servo motor-driven wiper system, ensuring clear visibility with minimal delay .

Below table 8 shows "Time per sweep and speed of wiper for different rain intensities

Table 8: Time per sweep and speed of wiper for different rain intensities

Rain Rate	Voltage	Time Per Sweep(s)	Speed (m/s)
Light drizzle	3.2	10.0	0.0314
Light Rain	2.6	9.0	0.0349
Moderate Rain	2.0	8.0	0.0393
Steady rain	1.4	7.0	0.0419
Heavy rain	0.8	6.0	0.0523

The table 8 illustrates how varying rain rates influence the voltage output of a rain sensor and correspondingly affect the wiper speed. As rain intensity increases from light drizzle to heavy rain, the voltage decreases, prompting faster wiper operation with shorter sweep times and higher wiper speeds for effective clearing.

2. Response Time

- The system responded within **5 m/s**.

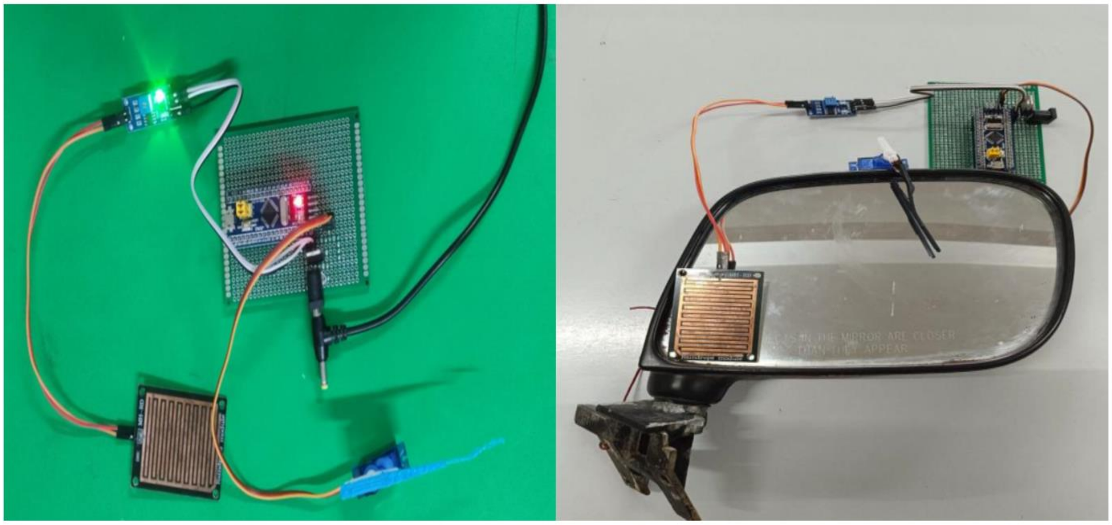

Figure 6 shows the hardware setup and real time mirror attachment

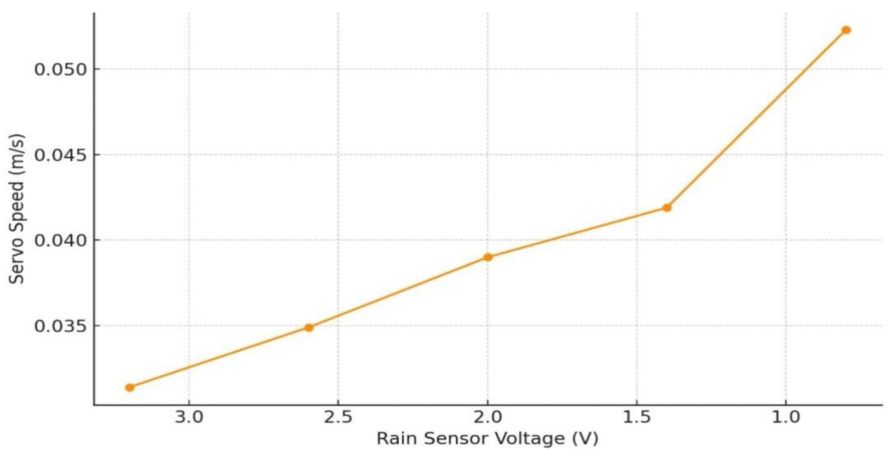

Figure 7. Change in speed of wiper vs output voltage

Figure 7 shows the graph of change in speed of wiper vs output voltage . As the rain intensity increases, the sensor voltage decreases, and the wiper speed increases. This inverse relationship ensures that heavier rainfall triggers faster wiper movement, enabling more efficient water clearing from the mirror surface.

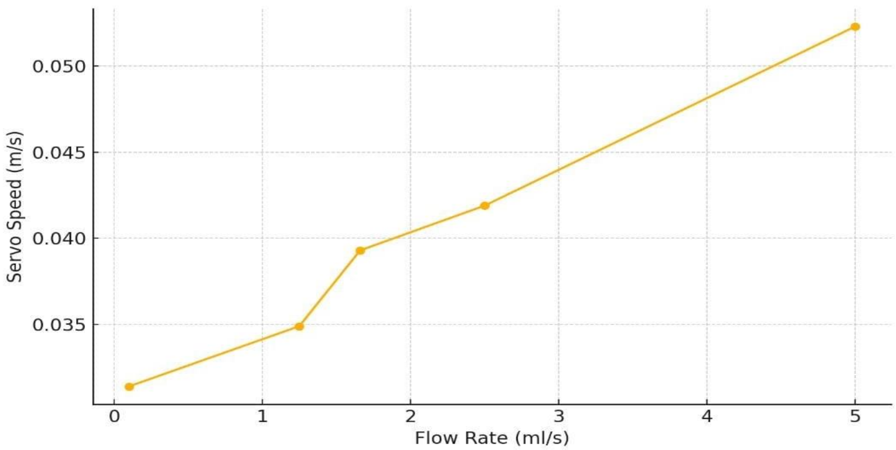

Figure.8 Rain Fall (ml) vs change in servo speed

Above figure 8 shows a direct relationship between rainfall (flow rate in ml/s) and servo speed. As the flow rate increases, indicating heavier rainfall, the servo (wiper) speed also increases, ensuring quicker and more efficient water removal from the mirror surface.

4. Scalability and Reliability: The system is designed to function reliably under diverse environmental conditions such as rain, mist, or dust. Its modular design ensures easy scalability and adaptability to different vehicle models with minimal hardware adjustments.

CHAPTER 10:

ADVANTAGES, DISADVANTAGES & APPLICATIONS

10.1 ADVANTAGES

1. **Automatic Rain Detection and Response:**The system automatically detects rain using a sensor and activates the wiper without driver input, improving safety and convenience.
2. **Enhanced Visibility:** Maintains a clear view of the side mirror during rainfall, helping the driver maintain awareness of surrounding vehicles and obstacles.
3. **Low Power Consumption:**Uses energy-efficient components like STM32 and SG90 servo motor, making it ideal for integration into vehicle electronics.
4. **Compact Design:** Lightweight and space-saving design allows easy installation on various vehicle models without affecting aesthetics.
5. **Fast Response Time:** The system reacts instantly to rainfall detection, ensuring immediate cleaning of the mirror surface.
6. **Weather-Resistant Operation:** Designed to function reliably in outdoor automotive conditions, including varying temperatures and humidity levels
7. **Cost-Effective Solution:** Utilizes affordable electronic components, providing a low-cost solution to improve driving safety.
8. **Customizable Operation:** Parameters like wipe interval and motor speed can be adjusted via programming, allowing system tuning per vehicle type**.**

10.2 DISADVANTAGES

1. **Sensor Sensitivity Limitations:** The YL-83 rain sensor may sometimes fail to detect light mist or rapidly evaporating droplets, which can reduce the system's responsiveness in mild weather conditions.
2. **Servo Motor Load Constraints:** The servo motor used may not be suitable for prolonged operation under heavy rain or dirt buildup on the mirror, potentially leading to mechanical wear or performance degradation
3. **Power Dependency:** The system depends on a stable power supply via voltage regulation (e.g., 7812). In the event of battery fluctuations or failure, the wiper system may stop functioning altogether.

4. **Installation Complexity :** The integration of STM32 microcontroller, sensor modules, and motor systems requires careful setup and wiring, making it less suitable for users without technical expertise.

10.3 APPLICATIONS

1. Automotive Safety Systems:

- Enhances driver visibility during rain by automatically clearing the side mirrors.
- Reduces the need for manual mirror wiping, allowing better focus on the road.
- Improves safety during night driving or lane changes in wet weather.

1. **Smart Vehicle Integration:**

- Can be integrated with modern vehicles' onboard microcontrollers and rain sensors.
- Adds automation to existing safety features, complementing ADAS (Advanced Driver Assistance Systems).
- Compatible with smart control units for remote or app-based activation.

2. **Public Transportation Vehicles:**

- Useful in buses and trucks, where side mirrors are large and more prone to water accumulation.
- Ensures consistent mirror visibility, especially during long-distance or all-weather operations.

3. **Automotive Research & Development:**

- Serves as a prototype for testing smart mirror technologies.
- Provides a base for experimenting with AI-based rain intensity detection.
- Contributes to innovation in autonomous and semi-autonomous vehicle systems.

CHAPTER 11:

CONCLUSION & FUTURE SCOPE

11.1 CONCLUSION

This article presents a practical, cost-effective, and intelligent solution for automated side mirror cleaning using a rain sensor integrated with an STM32 microcontroller and a servo motor. By continuously monitoring environmental conditions and responding instantly to rain or moisture detection, the system ensures clear visibility on vehicle side mirrors without manual intervention.

The primary strength of the system lies in its real-time responsiveness and compact hardware design. The integration of smart sensing with precise servo motor control allows for reliable and efficient wiping action only when necessary, minimizing mechanical wear and power consumption. The solution can be easily adapted to various vehicle types, supporting scalability and broader implementation.

Overall, the project demonstrates a robust and user-friendly approach to automated mirror maintenance. It effectively addresses key concerns such as weather adaptability, safety enhancement, and system autonomy—establishing a foundation for future upgrades such as rain intensity-based wiping speed, wireless diagnostics, and integration with broader vehicle safety systems.

During experimental testing, the system's sweeping mechanism was evaluated under varying rain intensities. The arc length of the wiper motion remained approximately constant (0.314m) throughout the tests. As rain intensity increased, the accumulation time on the rain sensor decreased, leading to a corresponding drop in the sensor's output voltage. At a low rain intensity (3.2 V output), the sweep time was recorded as 10.0 seconds, resulting in a speed of 0.0314 m/s. With increasing rain intensity and a sensor output of 2.6 V, the sweep time reduced to 9.0 seconds, and the speed increased to 0.0349 m/s. At 2.0 V, the sweep time was 8.0 seconds with a speed of 0.0393 m/s.

Further increase in rain intensity resulted in a 1.4 V output, reducing sweep time to 7.0 seconds and increasing speed to 0.0419 m/s. At the highest intensity, the sensor output dropped to 0.8 V, sweep time decreased to 6.0 seconds, and the speed reached 0.0523 m/s.

11.2 FUTURE SCOPE

This research envisions several promising advancements for enhancing and expanding the automated side mirror wiper system. As technology progresses, various improvements can be explored to optimize system performance, increase efficiency, and extend its applicability.

Integrating additional sensors such as humidity, temperature, and wind speed sensors can enhance the system’s adaptability to varying environmental conditions, ensuring precise and efficient operation. Additionally, incorporating wireless communication technologies like Bluetooth, Wi-Fi, or LoRa will enable remote monitoring and control, improving user convenience and accessibility.

The use of artificial intelligence (AI) and machine learning algorithms can introduce predictive analytics and adaptive control, allowing the system to anticipate rainfall intensity, optimize wiper speed, and enhance performance. AI-driven automation will contribute to better energy efficiency and minimize unnecessary wiper activation.

Another promising future development is the integration of mobile applications, enabling users to remotely monitor and adjust the system’s settings. Real-time alerts and notifications can enhance system responsiveness and reliability, providing users with timely updates on weather conditions and system status.

Furthermore, leveraging IoT-based cloud platforms can facilitate large-scale data collection and analysis, offering insights into weather patterns and improving predictive maintenance strategies. Cloud integration will enable seamless updates and optimization of the wiper system based on real-world data.

To enhance sustainability, the system can be designed to operate on energy-efficient components or renewable energy sources such as solar power, making it suitable for electric and hybrid vehicles. This approach will contribute to eco-friendly and low-power vehicle solutions.

CHAPTER 12:

REFERENCES

REFERENCES

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CHAPTER 13:

BILL OF MATERIAL

BILL OF MATERIAL

SR. NO	COMPONENT NAME	QUANTITY	UNIT COST (RS)	TOTAL COST
1	STM 32 Board	1	150	150
2	ST link v2 programmer	1	150	150
3	Rain Sensor	1	50	50
4	L7805 Voltage Regulator	1	15	15
5	SG90 Servo Motor	1	60	60
6	Power Jack	1	15	15
7	Female-Female Header	1	15	15
8	Wires	10	3	30
9	3D Printing	1	200	200
	Total Cost			**685**

Table 9: Bill of Material

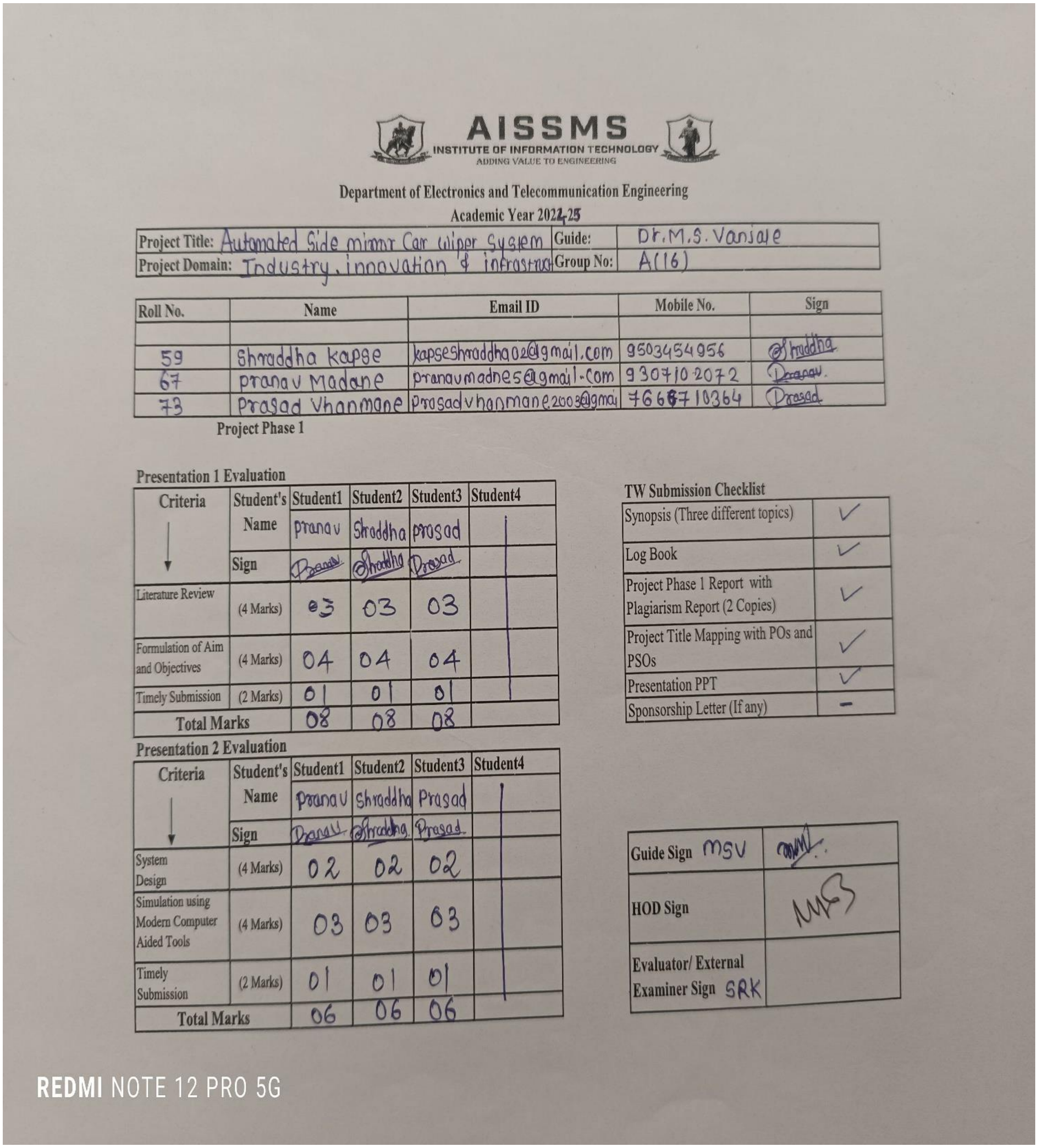
s

CHAPTER 14:

APPENDIX

14.1 PROJECT EVALUATION

PHASE 1



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Department of Electronics and Telecommunication Engineering
Academic Year 2024-25

Project Title: Automated Side mirror Car wiper System		Guide:	Dr. M.S. Vanjale
Project Domain: Industry, innovation & infrastruct		Group No:	A(16)

Roll No.	Name	Email ID	Mobile No.	Sign
59	Shraddha Kapse	kapseshraddha02@gmail.com	9503454956	Shraddha
67	Pranav Madane	pranavmadnes@gmail.com	9307102072	Pranav.
73	Prasad Vhanmane	Prasadvhanmane2003@gmai	7666710364	Prasad

Project Phase 1

Presentation 1 Evaluation

Criteria		Student1	Student2	Student3	Student4
	Student's Name	Pranav	Shraddha	Prasad	
	Sign	Pranav	Shraddha	Prasad	
Literature Review	(4 Marks)	03	03	03	
Formulation of Aim and Objectives	(4 Marks)	04	04	04	
Timely Submission	(2 Marks)	01	01	01	
Total Marks		08	08	08	

TW Submission Checklist

Synopsis (Three different topics)	✓
Log Book	✓
Project Phase 1 Report with Plagiarism Report (2 Copies)	✓
Project Title Mapping with POs and PSOs	✓
Presentation PPT	✓
Sponsorship Letter (If any)	–

Presentation 2 Evaluation

Criteria		Student1	Student2	Student3	Student4
	Student's Name	Pranav	Shraddha	Prasad	
	Sign	Pranav	Shraddha	Prasad	
System Design	(4 Marks)	02	02	02	
Simulation using Modern Computer Aided Tools	(4 Marks)	03	03	03	
Timely Submission	(2 Marks)	01	01	01	
Total Marks		06	06	06	

Guide Sign MSV	
HOD Sign	
Evaluator/ External Examiner Sign SRK	

REDMI NOTE 12 PRO 5G

PHASE 2

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Department of Electronics and Telecommunication Engineering
Academic Year 2022-23

Project Phase 2

Presentation 3 Evaluation

Criteria		Student1	Student2	Student3	Student4
	Student's Name	Shraddha	pranav	prasad	
	Sign				
System Implementation	(4 Marks)	02	02	02	
Working Model	(4 Marks)	02	02	02	
Timely Submission	(2 Marks)	02	02	02	
Total Marks					

Final Project Submission Checklist

Project Abstract	✓
Plagiarism report (Project Report and Paper)	✓
Published paper in UGC/WoS/Scopus care journal with DOI	-
Two signed copies of report with CD	-
Participation Certificate (Project or Paper Presentation Competition)	-
Project Hardware	✓
Project group photo with Guide (Soft and hardcopy)	✓

Presentation 4 Evaluation

Criteria		Student1	Student2	Student3	Student4
	Student's Name	Shraddha	pranav	prasad	
	Sign				
Working Model with Enclosure	(3 Marks)	02	02	02	
Effective Presentation	(3 Marks)	02	02	02	
Project Report	(3 Marks)	02	02	02	
Timely Submission	(1 Marks)	01	01	01	
Total Marks					

Guide Sign	
HOD Sign	
Evaluator /External Examiner Sign (SRK)	

Paper publication/IPR Details:

Paper Title:		Journal/Conference Name:	
Authors:			
DOI/IPR No.			

14.2 ABSTRACT

ONE PAGE SYNOPSIS

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Department of Electronics and Telecommunication Engineering

Automated side Mirror wiper system
(GOAL 9: Industry, Innovation and Infrastructure)

Division-BE(A)	**Group Number**-A(16)

1. **Shraddha Manmathappa Kapse (59)**
2. **Pranav Satish Madane (67)**
3. **Prasad Pandurang Vhanmane (73)**

Name of Internal Guide – Dr. M. S .Vanjale

Problem Statement - Design and Develope automated wiper using rain sensor and motor driver to rotate the wiper using microcontroller STM 32

Project Domain :

Objectives - **1.** Improved Visibility.
2. Increased Safety.
3. Cost-effectiveness.

Abstract – abstact In rainy conditions, maintaining clear visibility through side mirror is essential for safe driving, especially when checking blind spots or turning. However, raindrops and water streaks can obscure a driver's view, creating potential hazards. To address this, the "Rain-Sensing Side Window Wiper System" provides an innovative solution by automatically cleaning the side windows during rain.

The "Rain-Sensing Side Mirror Wiper System" is particularly useful in heavy rain or when vehicles are moving at slower speeds, where natural wind flow may not be sufficient to clear the windows

Project photo including Group members and guide during Presentation (Geo Tag Photo).

Project Mapping with Program Outcomes (1 – Slight, 2- Moderate, 3- Substantial)

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
3	3	3	3	3	2	3	3	3	3	3	3	3	3sn

Participation in Competition: Give the Competition Details

Awards (if any):

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Department of Electronics and Telecommunication Engineering

BE Project/ Academic Year 2024-25

Self-Evaluation Sheet 1

Group Number:

Project Title: Automated Side Mirror Car Wiper System .

1. Shraddha Kapse
2. Pranav Madane
3. Prasad vhanmane

Literature Survey (5)	Design (20)	Implementation (20)	Test Result (5)	Attendance on the Project Day (10)	Working according to plan activity (5)	Maintaining Log Book (5)	Research Paper (5)	Participation in Project/Poster Competition (5)	Award/ Prize if any (5)

Observation and Comments of Guide:

____

Student Name: Sign

Shraddha Kapse

Pranav Madane

Prasad Vhanmane

Guide Name: M.S.Vanjale Sign

Department of Electronics and Telecommunication Engineering

BE Project/ Academic Year 2024-25

Self-Evaluation Sheet 2

Group Number:	**A16**
Project Title:	**Automated Side Mirror Car Wiper System**
Date:	

1. Describe your feelings about working on the project. Did you enjoy working on it?

ANS: Working on this project was a genuinely rewarding experience. It gave me an opportunity to apply what I had learned in my coursework to a real-world application. I enjoyed the process of designing, experimenting, and troubleshooting the system, and it felt satisfying to see our idea take shape into a functional prototype.

2. What was the hardest part about working on your project?

ANS: The most challenging part of the project was ensuring the wiper system responded accurately to water detection on the side mirror. Fine-tuning the sensor sensitivity and timing the motor activation for optimal performance required extensive testing and adjustments. We also encountered difficulties in maintaining consistent operation during varying weather conditions and integrating the system without interfering with the car's existing electronics. Overcoming these issues involved a lot of troubleshooting and iterative improvements.

3. List some of the things you learnt while working on your project?

ANS: Throughout the project, I learned how to interface rain and moisture sensors with a microcontroller to automate mechanical systems. I gained practical experience in working with Dservo motors, relay, and control circuits. I also improved my skills in embedded C programming, system integration, and debugging sensor-based responses. Additionally, I learned how to manage power supply requirements, ensure reliable waterproofing for outdoor components, and collaborate effectively within a team setting to meet project goals.

4. Were you satisfied with your final project?

ANS: Yes, I was quite satisfied with the final outcome of our project. The system worked according to our expectations, and we were able to demonstrate Automated Side Mirror with good accuracy and responsiveness. Although there were a few minor improvements that could be made, the overall performance was stable, and the results were encouraging.

5. What did you like the best about your project?

ANS: The best part of the project for me was seeing the automated response of the side mirror wiper to real-time weather conditions. It was very satisfying to watch the system detect rain or moisture and activate the wiper without any manual input. This demonstrated the effectiveness of our sensor integration and control logic

6. What did you like the least about the project?

ANS: The part I liked the least was dealing with the initial setup issues and debugging errors that weren't easy to trace. Sometimes the system would not respond as expected, and finding out whether the problem was in the code, wiring, or hardware was time-consuming and frustrating. However, these challenges were part of the learning experience.

7. What was the most important thing that you learnt from doing your project that will help you in the future?

ANS: The most important thing I learned was how crucial it is to plan, integrate, and test every part of the system carefully. This project taught me the importance of real-time decision-making in embedded systems and how even small inaccuracies can affect the performance of the entire system.

Student Name: Sign

Shraddha Kapse

Pranav Madane

Prasad Vhanmane

Guide Name: M.S..Vanjale Sign

PLAGIARISM REPORT

iThenticate Page 2 of 77 - Integrity Overview Submission ID trn:oid:::3618:100752164

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DEPARTMENT E&TC ENGINEERING

Project Title Mapping with PO-PSO-MAPPINNG:

Project Mapping with Program Outcomes (1 – Slight, 2- Moderate, 3-Substantial)

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
3	3	3	3	3	2	3	3	3	3	3	3	3	3

Justification:

PO1: The knowledge we had gained so far was utilized to convert a conventional flow control process into an automated system using Rain sensors. This innovation aimed to make industrial operations more efficient and accurate.

3/3

PO2: We reviewed various technical papers and research studies to identify existing challenges in manual and semi-automated flow systems. By analyzing those complex problems, we were able to design a system that addressed the core issues and offered potential improvements.

3/3

PO3: Initially, the design involving sensor integration and flow control logic was complex. With the help of technical resources and expert guidance, we simplified the system for real-world application while keeping safety and industrial standards in check.

3/3

PO4: We studied the experiences of industries facing difficulties in manual flow control. Using that input, we selected appropriate sensors and automated mechanisms to create an efficient and reliable solution tailored to those operational needs.

3/3

PO5: Before developing our hardware prototype, we created and tested the system through simulations using tools like Proteus. This helped us anticipate faults and refine the working model before moving on to the actual circuit and component integration.

3/3

PO6: Our project ensures ease of use for non-technical users in industrial environments. By pressing a few keys or initiating minimal input, users can control the entire system, which encourages wider adoption of simple automation technologies.

2/3

PO7: The system eliminates the need for constant human monitoring and physical interaction with flow valves or counters. It promotes automation in industrial setups, aiming to save time, reduce errors, and improve efficiency.

3/3

PO8: We understood our responsibility as engineering students to offer practical and safe technological solutions. This project emphasizes secure and efficient flow management, especially in areas where manual control may be prone to risks.

3/3

PO9: Teamwork played a key role in this project. We shared responsibilities like coding, simulation, hardware testing, and documentation. By understanding each other's capabilities, we supported one another to bring out the best results.

3/3

PO10: We interacted with domain experts to validate our approach and incorporated their suggestions in our prototype. We also documented our findings, prepared technical reports, and created presentations to communicate our innovation within the academic and industrial circles.

3/3

PO11: Project planning and resource management were crucial. We had to stay within a budget while selecting reliable components and managing time effectively, which helped us achieve our goals efficiently and economically.

3/3

PO12: This project taught us that learning is an ongoing process. As engineering and sensor technology keep evolving, we realized the importance of updating ourselves regularly to build smarter, more sustainable solutions in the future.

3/3

PSO1: We applied knowledge of embedded systems, microcontrollers, sensor integration, and programming using Embedded C to develop the automated flow controller. These skills helped in building a system that is accurate and responsive.

3/3

PSO2: Simulation tools like Proteus helped us verify the working of the sensor and controller logic before physical implementation. We then used the simulation results to optimize the actual circuit.

3/3

Project Outcome:

The project successfully led to the development of a smart and automatic side mirror wiper system using the STM32 microcontroller, YL-83 rain sensor, and servo motor. The system responds efficiently to the presence of water on the mirror surface, enhancing driver visibility and safety.